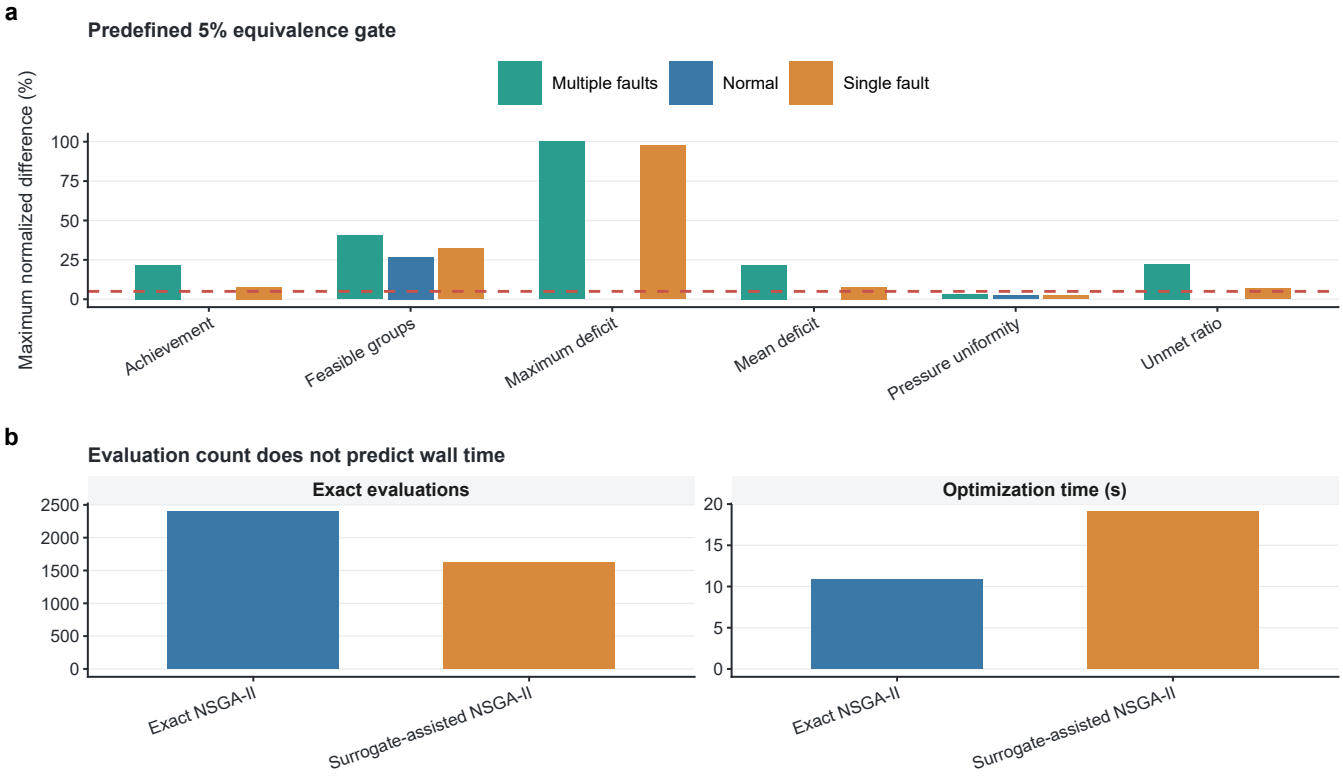
a
Predefined 5% equivalence gate
Multiple faults
Normal
Single fault
Maximum normalized difference (%)
100
75
50
25
0
Achievement
Feasible groups
Maximum deficit
Mean deficit
Pressure uniformity
Unmet ratio
b
Evaluation count does not predict wall time
Exact evaluations
2500
2000
1500
1000
500
0
Exact NSGA-II
Surrogate-assisted NSGA-II
Optimization time (s)
20
15
10
5
0
Exact NSGA-II
Surrogate-assisted NSGA-II